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DATA SCIENCE

Logistic Regression



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Logistic Regression – Introduction

In Linear regression, the outcome variable is continuous and the predictor variables can be a mix of numeric and categorical. But often there are situations where we wish to evaluate the effects of multiple explanatory variables on a binary outcome variable

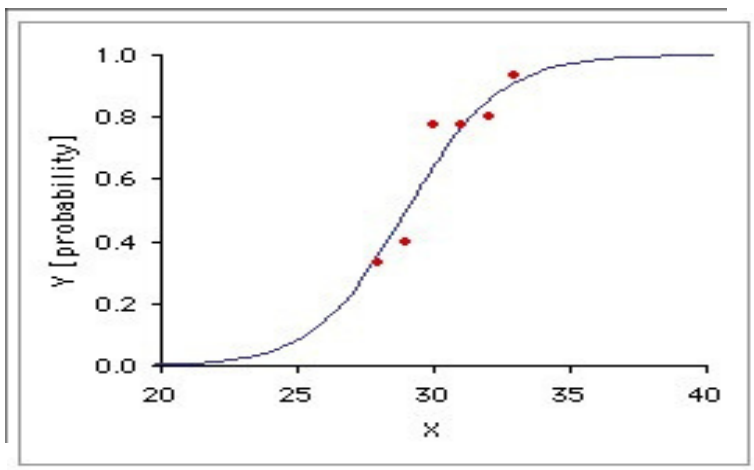
For example, the effects of a number of factors on the development or otherwise of a disease. A patient may be cured or not; a prospect may respond or not, should we grant a loan to particular person or not, etc.

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Logistic Regression – Introduction

When the outcome or dependent variable is binary, and we wish to measure the effects of several independent variables on it, we uses Logistic Regression

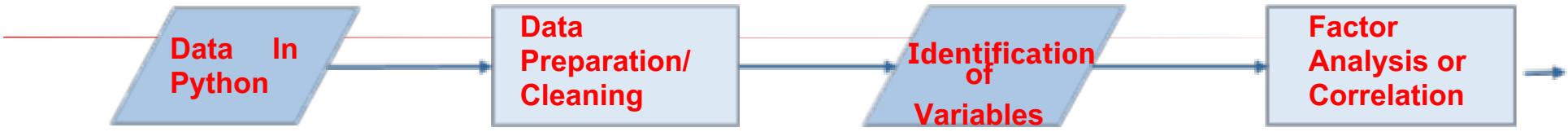
- The binary outcome variable can be coded as 0 or 1.
- The logistic curve is shown in the figure below:



We estimate the probability of success by the equation:

$$P = \frac{e^{a+bX}}{1+e^{a+bX}}$$

Process Flow

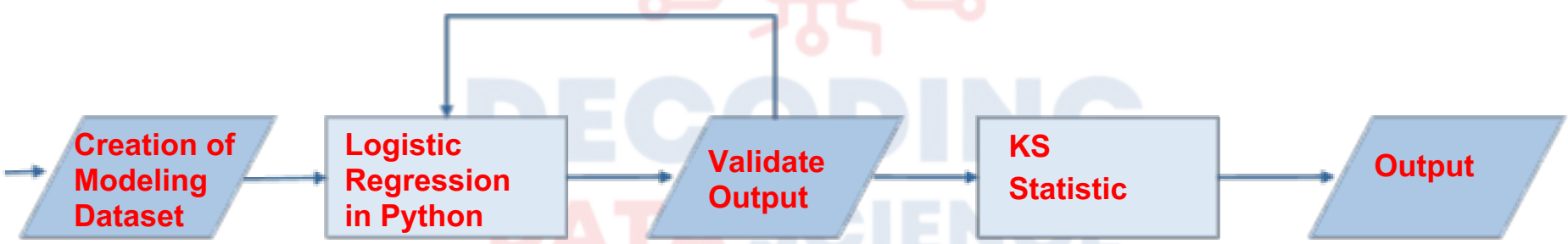


Data is obtained in pandas dataframe

- Missing Value
- Imputation Trash value
- Outlier Treatment

Independent and dependent variables should be identified

- FA is done in order to get the variables into groups
- Good to choose factor solution near the Eigen value of 1
- As a further Check Correlation Analysis is done



Divide data into Development and Validation Sample in ratio 70:30 or 80:20

- Assume all assumptions hold
- Check for the significance of the variables
- Run Regression on Development sample

Validate the output on the Validation sample ,by running the same model on the Validation sample

Results will be presented in PowerPoint

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Python code

Step 1: Importing the dataset

```
dataset = pd.read_csv('car_purchase_Ads.csv')
X = dataset.iloc[:, :-1].values
y = dataset.iloc[:, -1].values
```

Step 2: Splitting the dataset into the Training set and Test set

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.25, random_state = 0)
```

Step 3: Feature Scaling

```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

Python code

Step 4 : Training the Logistic Regression model on the Training set

```
from sklearn.linear_model import LogisticRegression
classifier = LogisticRegression(random_state = 0)
classifier.fit(X_train, y_train)
```

Step 5: Predicting a new result

```
print(classifier.predict(sc.transform([[30,87000]])))
```

Step 6: Predicting the Test set results

```
y_pred = classifier.predict(X_test)
print(np.concatenate((y_pred.reshape(len(y_pred),1), y_test.reshape(len(y_test),1)),1))
```

Python code

Step 7: Making the Confusion Matrix

```
from sklearn.metrics import confusion_matrix, accuracy_score
cm =confusion_matrix(y_test, y_pred)
print(cm)
accuracy_score(y_test, y_pred)
```

Practice

For location of code and dataset

https://github.com/arshad831/Modelling-Exercise/blob/main/logistic_regression.ipy nb

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- ✅ Ready to use Resume Template
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